

Asymptotic Notation

The **time complexity** of an algorithm is expressed in terms of the **number of basic operations** used by the algorithm for a particular size of input. The largest number of basic operations for a particular input size is the **worst case time complexity**.

Basic operations include additions, multiplications, comparisons, sums and assignments.

— f is $O(g)$ if there are constants C and K such that:

$$|f(x)| \leq C|g(x)| \text{ for all } x > K$$

Sum rule: if f_1 is $O(g_1)$ and f_2 is $O(g_2)$, then $f_1 + f_2$ is $O(\{\max\{|g_1(x)|, |g_2(x)|\}\})$

Product rule: if f_1 is $O(g_1)$ and f_2 is $O(g_2)$, then $f_1 \times f_2$ is $O(g_1(x) \cdot g_2(x))$

— f is $\Omega(g)$ if there are positive constants C and K such that:

$$|f(x)| \geq C|g(x)| \text{ for all } x > K$$

or equivalently, iff g is $O(f)$.

— f is $\Theta(g)$ if f is $O(g)$ and f is $\Omega(g)$.

— f is $o(g)$ if $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$

— f is $\omega(g)$ if g is $o(f)$, i.e. $\lim_{x \rightarrow \infty} \frac{g(x)}{f(x)} = 0$